



LAHORE WATER & WASTEWATER MANAGEMENT PROJECT - SEWERAGE SYSTEM FROM LARECHS COLONY TO GULSHAN-E- RAVI, LAHORE (THROUGH TRENCHLESS TECHNOLOGY) LAHORE WASTEWATER AND DRAINAGE PROJECT



PROJECT NAME

Sewerage system from LARECHS Colony to Gulshan-e-Ravi, Lahore (Through Trenchless Technology)

DEPARTMENT NAME/ SECTOR

Housing, Urban Development & Public Health Engineering Department / Water and Sanitation Agency.

FUNDING AGENCY

Asian Infrastructure Investment Bank (AIIB)

LOCATION

Laying of Trunk & Lateral Sewer Lines from LARECHS Colony & Gurumangat Road, to Gulshan-e-Ravi, Lahore.

PROJECT SCOPE

- Laying of Trunk Sewers through MTBM:
 - Line A:** LARECHS Colony to Qartaba Chowk, Lahore (incl. pipe dia of 1050, 1500 & 2400mm). Length = 12.14 km.
 - Line B:** Gurumangat to Qartaba Chowk, Lahore (incl. pipe dia of 1050, 1500, 2400mm). Length = 12.82 km.
 - Line C:** Qartaba Chowk to Gulshan-e-Ravi, Lahore (incl. pipe dia of 1050 & 3500 mm). Length = 7.25 km.
- Laying of Lateral Sewers (450 mm) through Open Cut. Length = 159 m
- Total Route Length = 32.37 km**
- Construction of Ultimate Disposal Station of Ultimate Capacity 848 cusecs.
- Rehabilitation of existing drains and collector channel.
- Laying of Rider Sewers (450, 600 & 750 mm) = 8.48 km approx.

APPROVED PROJECT COST

14,165 Million PKR

SERVING POPULATION AND AREA

2.35 Million and 15,658 Acres (approx.)

PROJECT DESCRIPTION

Micro-Tunneling Boring Machine (MTBM)

- The main trunk sewers and some laterals will be laid underground through Micro tunneling Boring Machine (MTBM).
- The MTBM of the following diameters for the mentioned lengths will be used:

#	Pipe Diameter (mm)	Length (km)
1	1050	9.837
2	1500	5.454
3	2400	11.582
4	3500	5.337
Total		32.210

- The number of jacking pits and receiving pits = 165 (approx.)
- The max spacing of jacking & receiving pits for different diameter of pipes:

#	Pipe Dia (mm)	Max Spacing (m)
1	1050	190
2	1500	250
3	2400	350
4	3500	500

Open-cut

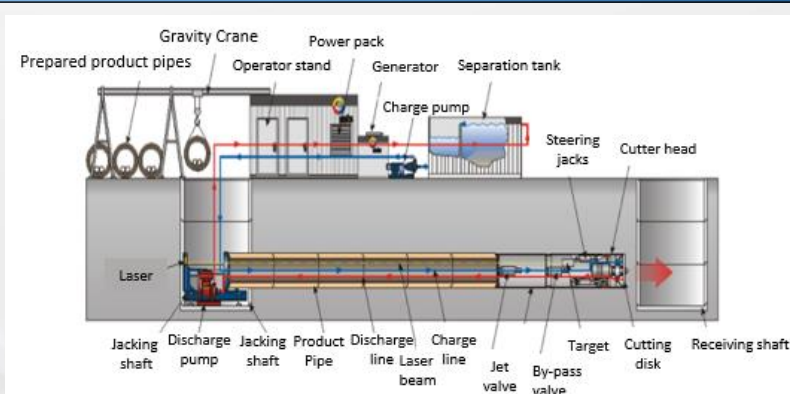
- The laterals & rider sewers will be laid by open-cut as per the following details:

#	Sewer	Pipe diameters (mm)	Length (km)
1	Laterals	450	0.159
2	Rider Sewers	450, 600, 750	8.48

MAIN COMPONENTS OF A MICRO TUNNELING PROCESS

The main components of micro tunnelling drive consist of:

- Micro-Tunnelling Drive/Machine
- Launching and Receiving shafts
- Gantry/Crane
- Slurry Loading Unit
- Generator
- MTBM Contractor Unit
- RCC with HDPE Lining Pipes (6000 psi strength)
- Dumping Truck



TIME FOR COMPLETION

- Design Phase-I = 6 Months
- Execution Phase II = 30 Months

MODE FOR EXECUTION

Engineering, Procurement and Construction (EPC) Mode – Design & Build

EXPECTED OUTCOMES

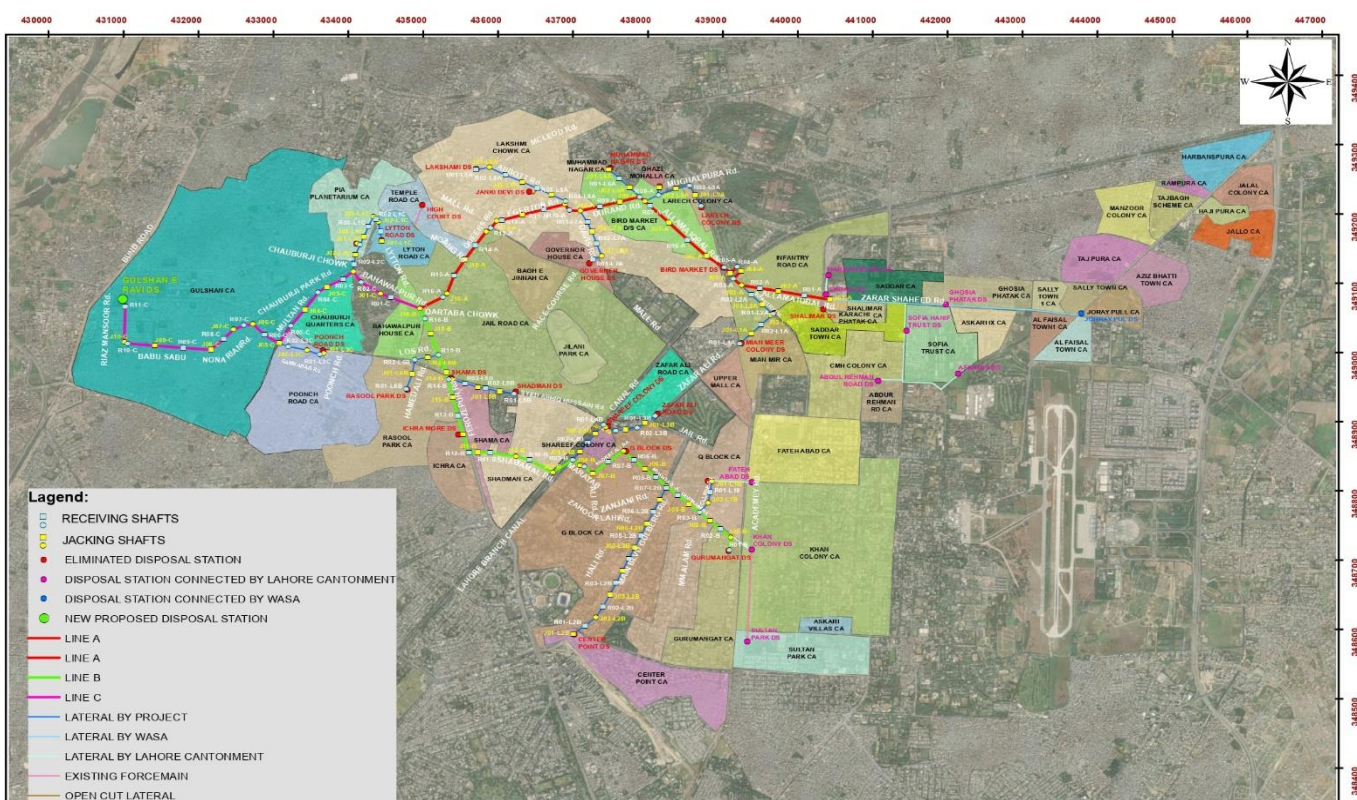
- Improvement in existing sewerage system by laying of new trunk sewers in Central Lahore.
- Safe disposal of sewage/ storm water from the project area.
- Environmental improvement in terms of restoration and rehabilitation of storm water channels which may be utilized for recreational activities within the city.

TASKS FOR DESIGN PHASE I

Deliverables

- 1 Feasibility Report
- 2 Updated Preliminary Engineering Design Report
- 3 Updated Geotechnical Report
- 4 Updated Topography Survey
- 5 Updated ESIA, LRP, RAP, GAP, SEP
- 6 Final tender Documents
- 7 Tender Evaluation Report
- 8 Award of Work

LOCATION FOR LAYING OF TRUNK SEWER



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WATER AND SANITATION
AGENCY, LAHORE



EBA
JSP
A Joint Venture

REVISION

REV	DATE	DESCRIPTION
04		
03		
02		
01		

CONTROL

DRAWN	
CHECKED BY	
RECOMMENDED	
CHOSEN	
APPROVED	

DATE

OCT 2024

DRAWING No.

FIGURE

REV:-

SCALE
1:50,000